

AI for Bharat Hackathon

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Team Name: Sudharshan AI 

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Problem Statement: Combatting the "Digital Arrest" epidemic and real-time UPI coercion using privacy-first, agentic financial intelligence.

Brief about the Idea:

The Crisis: India lost ₹1,750 crore to cyber fraud in H1 2024 alone, with 7.4 lakh complaints filed to NCRP. 85% of victims fail to act within the critical "Golden Hour"—the window before funds are withdrawn or laundered.

Sudarshan-AI is a real-time, on-device financial guardian that detects psychological coercion and fraud patterns during live UPI transactions. By combining Behavioral Biometrics with Agentic RAG, it prevents funds from leaving the victim's account during the "Golden Hour" of a crime, even if the user is being manipulated or isolated by scammers.

- **Differentiation:** Current systems are reactive (reporting after loss). Sudarshan is proactive (intervention before the COMMIT). Unlike bank filters that only see transaction amounts, we analyze the human state (voice pressure, typing tremors).
- **How it solves the problem:** It implements a "Silent Circuit Breaker"—holding funds in a secure line without alerting the scammer, giving law enforcement a required and reasonable tactical window to intervene.
- **USP:** "Zero-Knowledge" Safety. We analyze coercion signatures locally on the edge (no audio uploaded to cloud) to satisfy the **DPDP Act 2023**.

List of features offered by the solution

Core Features (Will demo):

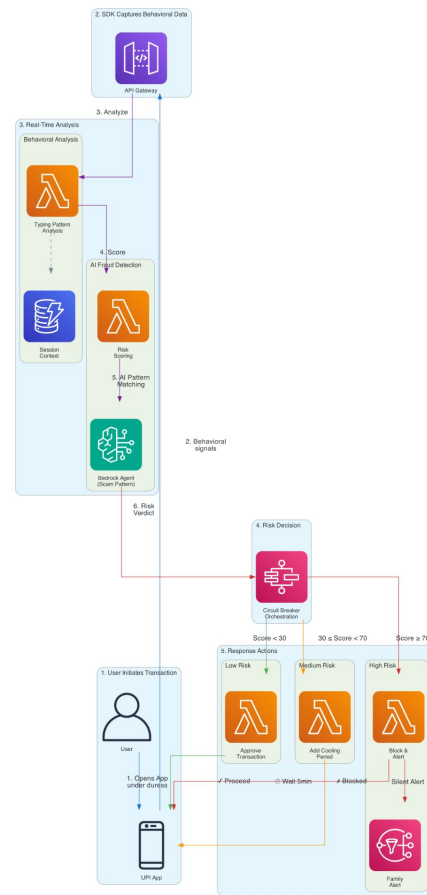
1. **Agentic Fraud-Script Analysis** - Bedrock Agents analyze transaction context against known scam patterns
2. **Silent Circuit Breaker** - Step Functions orchestrate fund hold without alerting scammer
3. **Behavioral Biometrics** - Typing patterns, navigation flow anomaly detection
4. **Emergency Duress PIN** - Secret PIN triggers silent alert to authorities

Roadmap Features (Future Vision):

1. Voice Sentiment Analysis (Phase 2 - Mobile SDK)
2. On-Device Edge AI for offline protection (Phase 3)
3. Nitro Enclaves for enterprise deployment (Phase 3)

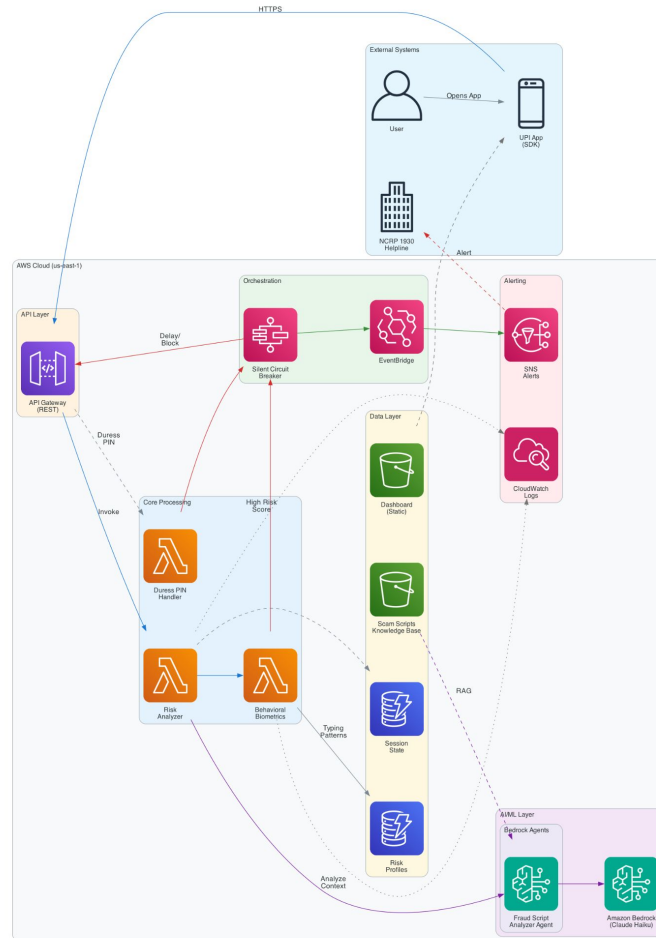
Process flow diagram:

1. User opens UPI app while on a scam call
2. SDK detects abnormal typing patterns (hesitation, tremors)
3. Bedrock Agent analyzes transaction context against scam script database
4. Risk score triggers Silent Circuit Breaker (Step Functions)
5. Transaction held + family/authorities silently alerted
6. Dashboard shows real-time risk visualization



Sudharshan-AI: Fraud Detection Flow

Architecture diagram of the proposed solution:



Sudharshan-AI: Real-Time UPI Fraud Prevention

Technologies & Implementation Costs in the solution:

Service	Purpose	Free Tier Charges
AWS Bedrock Agents (Haiku)	Fast, cheap classification	Pay-per-use (\$0.25/1M tokens)
AWS Bedrock Agents (Sonnet)	Complex agentic reasoning	Minimal usage for demos
AWS Lambda	Serverless compute	1M requests/month Free
Amazon DynamoDB	Transaction metadata, baselines	25GB Free
AWS Step Functions	Workflow orchestration	4K transitions/month Free
Amazon API Gateway	REST API endpoints	1M calls/month Free
Amazon S3	Static web hosting, logs	5GB Free
Amazon SNS	Alert notifications	1M publishes Free

Estimated Prototype Cost: \$15 - 30 (with Free Tier benefits)

Technical Differentiators:

Current Bank Systems	Sudarshan-AI
Detection: Rule-based: "Amount > X = Block"	Agentic AI: Reasons through scammer psychology
Reactive: Report after money lost	Proactive: Intervene before transaction commits
10-20 mins (manual helpline)	<1 second (automated intervention)
Invasive: Call recording, screen monitoring. Retrofitted for DPDP	Privacy-first: Behavioral signals only, no raw data stored. Built for DPDP Act 2023 from day one
Static rules, easily bypassed	Dynamic: Learns new scam patterns via Bedrock
English-centric interfaces	Designed for 250M+ non-English users

Business Impact & Feasibility:

- **Target Customers:** Banks, Fintechs (PhonePe, Google Pay, Paytm)
- **Business Model:** Safety-as-a-Service API - per-transaction micro-fee
- **Alignment:** IndiaAI Mission, RBI Advisory on real-time fraud prevention
- **Sustainability:** Reduces bank fraud losses → clear ROI for customers

Impact Metrics:

- **250M+** potential beneficiaries (non-English speaking UPI users)
- **Reduce MTTC** (Mean Time To Contain) from 10+ mins to <1 second
- **Potential:** To prevent ₹500+ crore annually based on current fraud volumes
- **Systemic impact:** Restore trust in Digital Public Infrastructure

Roadmap:

Phase	Timeline	Deliverable
Prototype	Feb 10-22, 2026	Core MVP with Bedrock + Step Functions
Phase 2	Q2 2026	Mobile SDK with voice sentiment
Phase 3		Edge AI for offline, Nitro for enterprise
Pilot	Q3 2026	B2B pilot with regional banks

Risk Mitigation:

- User can override with biometric + confirmation
- Trusted contacts whitelist for recurring payments
- Continuous learning reduces false positives over time

Conclusion:

> > Every day, thousands of Indians lose their life savings in seconds. Not because of weak passwords or stolen cards—but because scammers exploit something no firewall can protect: **human fear**. Current systems see transaction amounts. We see the trembling hands behind them.

National Alignment:

IndiaAI Mission: Citizen protection through sovereign AI

RBI Advisory 2025: Real-time fraud prevention mandate

DPDP Act 2023: Privacy-first by design

Digital Bharat: Built for 250M+ non-English speaking users

Our Commitment: We're not building another chatbot wrapper. We're building critical infrastructure for India's digital economy. Every UPI transaction should feel as safe as handing cash to a trusted neighbor.

Call to Action: With your support, we will transform this prototype into a production-ready safety layer that banks, fintechs, and government agencies can deploy to protect every Indian citizen in the digital economy.

Want to see the code: Here's the [repo](#)

The Tagline:

> > "We don't just detect stolen cards. We detect stolen minds."

— Sudarshan-AI: The Silent Guardian for Digital Bharat

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Thank You

